F78AB – Actuarial and Financial Mathematics
Computer Lab 3

Question 1:

Set up an Excel spreadsheet to compute the complete loan schedule for the following two loans. Remember to adjust the final repayment, if necessary, so that the loan is repaid exactly.

Loan A: A loan of £120 000 for a term of 25 years to buy a one-bedroom flat. Repayments are level and made annually in arrears. The rate of interest is 4% per annum effective.

Loan B: A loan of £20 000 for a term of 5 years to buy a car. Repayments are level and made monthly in arrears. The rate of interest is 4% p.a. effective.

You should **round the appropriate values** in your schedule to 2 decimal places. This is not the same as formatting the number displayed to 2 decimal places.

Question 2:

Suppose that after two years have passed, the lender changes the rate of interest to 5% per annum effective. Using a new worksheet in your workbook, amend the two loan schedules accordingly.

Question 3:

Consider only Loan A.

After three years have passed since you took out the loan, you ask the lender if you can extend the term of the loan by an extra three years. The lender agrees, but only if the interest rate is raised to 5.5% per annum effective.

Using a new worksheet in your spreadsheet, amend the loan schedule accordingly.

For Questions 4 and 5 below, you will need to use Goal Seek and you should **NOT round the values** in your schedule.

Question 4:

Use Excel to find the repayment amount in **example 7.6.1** from the notes.

Question 5:

Do the calculations for **Tutorial 5 Question 7.1** in Excel.